for some function $f\colon G \times G \to M$. If $(M, \diamond_M)$ and $(G, \diamond_G)$ already satisfy a law E, then this extension will also satisfy E if and only if $f$ satisfies a certain "cocycle equation", which is a linear equation on $f$. One can then sometimes use linear algebra to locate an $f$ that satisfies the cocycle equation for one law E but not another E′, thus refuting the implication E $\models$ E′. An example is as follows:

**Proposition 5.15** (E1110 $\not\models$ E1629)**.** *The law* $\mathrm{x} \simeq \mathrm{y} \diamond ((\mathrm{y} \diamond (\mathrm{x} \diamond \mathrm{x})) \diamond \mathrm{y})$ (E1110) *does not imply* $\mathrm{x} \simeq (\mathrm{x} \diamond \mathrm{x}) \diamond ((\mathrm{x} \diamond \mathrm{x}) \diamond \mathrm{x})$ (E1629)*, even for finite magmas.*

*Proof sketch.* Using the linear ansatz, we find that E1110 has a model $\mathcal{M}$ with carrier $\mathbb{F}_5$ (the finite field $\mathbb{Z}/5\mathbb{Z}$) with operation $x \diamond y = 3x - y$. We then apply the ansatz (8) with $G = M$. One then finds that this operation satisfies E1110 if $f\colon \mathbb{F}_5 \times \mathbb{F}_5 \to \mathbb{F}_5$ the cocycle equation

$$3f(x,x) - 3f(y,2x) - f(3y-2x,y) + f(y,3y-x) = 0$$

for all $x, y \in \mathbb{F}_5$, and satisfies E1629 if $f$ satisfies the cocycle equation

$$f(2x,0) - f(2x,2x) = 0$$

for all $x \in \mathbb{F}_5$. A routine symbolic algebra package computation reveals that the space of $f$ that satisfies the former equation is a six-dimensional vector space over $\mathbb{F}_5$, which is not contained in the solution space of the latter equation, giving the claim. In fact, since these equations preserve the space of homogeneous polynomials of a fixed degree, one can use linear algebra to locate an example that is a homogeneous polynomial; one explicit choice is

$$f(x,y) = y^5 + xy^4 + x^2y^3 + 3x^3y^2 + 3x^4y. \qquad \square$$

It may be of interest to develop this theory of "magma cohomology" further, for instance by defining higher order magma cohomology groups.

Now we give an example of how method (ii) can be combined with method (i).

**Proposition 5.16** (E1659 $\not\models$ E4315)**.** *The law* $\mathrm{x} \simeq (\mathrm{x} \diamond \mathrm{y}) \diamond ((\mathrm{y} \diamond \mathrm{y}) \diamond \mathrm{z})$ (E1659) *does not imply* $\mathrm{x} \diamond (\mathrm{y} \diamond \mathrm{x}) \simeq \mathrm{x} \diamond (\mathrm{y} \diamond \mathrm{z})$ (E4315).

*Proof.* There are two simple models for E1659: the model $G$ with carrier $\mathbb{Z}/2\mathbb{Z}$ and operation $x \diamond y = x + 1$, and the model $\mathcal{M}$ with carrier $\mathbb{Z}$ and operation $x \diamond y = x$. Using the ansatz (8), one can soon discover that one obtains a magma operation $\diamond : (G \times \mathbb{Z}) \times (G \times \mathbb{Z}) \to G \times \mathbb{Z}$ with $f(0,0) = f(1,0) = 0$, $f(0,1) = -1$, and $f(1,1) = 1$. This model still satisfies E4315. However, we can create a modification $\diamond'$ of $\diamond$ as follows. We will seek to violate E4315 at $x = (0,0)$, $y = (0,0)$, $z = (1,0)$, thus we want

$$(0,0) \diamond' ((0,0) \diamond' (0,0)) \neq (0,0) \diamond' ((0,0) \diamond' (1,0)).$$

We have $(0,0) \diamond (0,0) = (1,0)$ and $(0,0) \diamond (1,0) = (1,-1)$. One can try to force the counterexample by setting $(0,0) \diamond' (1,0)$ to equal $(0,0)$ instead of $(1,-1)$. However, if this is the only change we make, then we no longer satisfy E1659, since

$$(1,0) \neq ((0,0) \diamond' (1,0)) \diamond' (((1,0) \diamond' (1,0)) \diamond' (1,t))$$

for any $t \in \mathbb{N} \backslash \{0\}$. But these are the only counterexamples created involving elements in the subset $G \times \mathbb{N}$ of $G \times \mathbb{Z}$; and if one then sets $(0,0) \diamond' (1,t) = (0,0)$ for *all* $t \in \mathbb{N}$, and

then restricts to $G \times \mathbb{N}$ (which is now closed under $\diamond'$), then one can check that the modified operation $\diamond'$ on this submagma now satisfies E1659 but not E4315 as required. Incidentally, this submagma is isomorphic to the magma $\mathcal{M}' = (\mathbb{Z}, \diamond'')$ with $m \diamond'' n = -m$ for $n < 0$ and $m \diamond'' n = -m - 1$ for $n \geq 0$ under the bijection that maps $(0, s)$ to $s$ and $(1, s)$ to $-s - 1$. $\square$

The specific law $\mathrm{x} \simeq \mathrm{x} \diamond ((\mathrm{y} \diamond \mathrm{z}) \diamond (\mathrm{x} \diamond \mathrm{z}))$ (E854) turned out to be somewhat "mutable", in the sense that one can often change a small number of entries in the multiplication table of a finite magma satisfying this law, or add rows and columns to the table, in ways that preserve the law E854. This makes the law amenable to methods (i) and (iii) to construct new models of this equation that refute various implications E854 $\not\models_{\text{fin}}$ E, for instance by starting with a model that already refuted some stronger law E′, and then attempting to modify it (possibly with ATP assistance) by some combination of methods (i) and (iii) to produce a model that violates E.

Some heuristics loosely inspired by discrete-time dynamical systems proved helpful. The idea is to generate a sequence of magmas, each of which is generated from the previous entry by applying various operations expected to increase the likelihood of finding a refutation. This is similar to the greedy methods in Section 5.5, except that we require our resulting magma to be finite and completely defined, and our transformations need not be deterministic.

Since our goal is simply to find a finite model — and any candidate can be checked directly — we are not limited to transformations that can be rigorously justified. SAT solvers like *Glucose* [3, 4] inspired by the earlier *MiniSAT* [25], via convenient interfaces like *PySAT* [30, 31], other fast SAT solvers such as *Kissat* [12], counterexample finders like *Mace4* [43], and more general ATPs like *Vampire* [38] which can be used as solvers, all succeeded in finding useful magmas following this approach.

For example, suppose our goal is to show E854 $\not\models_{\text{fin}}$ E, for some E. We start with a magma table which satisfies both equations, and remove a random subset of the table entries (creating a partially defined magma). We then ask the ATP to find a magma which satisfies E854, filling in the unspecified values and potentially adding new elements. If in fact E is not implied by E854, the ATP might succeed in finding a magma refuting the implication. We may also directly insert a violation of E into the magma cells we have emptied, hoping that a consistent completion still exists. Another method, by analogy with slowly introducing forcing terms into numerical integrations while preserving adiabatic invariants, is to impose selected implications of a given equation without directly enforcing the equation itself. This can gradually drive the magmas in the sequence toward satisfying the equation without immediately imposing it.

Combining several of these techniques allowed us to find proofs of

$$\text{E854} \not\models_{\text{fin}} \text{E413}, \text{E1045}, \text{E1055}, \text{E3316}, \text{E3925}.$$

As an example of the process, we started with a magma that satisfied E854 and all the above equations, but had an E10 violation. We suspected that E10 might be relevant (and later found an argument showing multiple E10 violations would be required in a magma which showed E854 $\not\models_{\text{fin}}$ E1055), and so used *Kissat* and *Vampire* to grow E854 magmas with such violations. These larger magmas were then used as seeds for the random evolution process,

and the ATPs eventually succeeded in finding magmas which satisfied E854 but refuted the implications in question.

Note that these approaches have their limitations. To be effective, it must be easy to transition between different states, which usually involves finding a magma satisfying the equation of interest or at least some related one. Equations whose finite magmas are difficult to find (e.g. E677), whether because of absolute rarity or the numerical challenges that our ATPs have in finding them without taking advantage of a structural ansatz, appear resistant to these methods in practice.

Another way to utilize (iii), which proved useful for laws that involved the squaring operator $S$, was to adopt a "squares first" approach in which one selected a base model $S\mathcal{M} = (SM, \diamond)$ to serve as the set of squares, then extend it to a larger model $\mathcal{M}$ with carrier $M = SM \uplus N$ by first determining what the multiplication map should be on the diagonal $\{(x, x) : x \in N\}$ (i.e., to determine the squaring map $S\colon N \to SM$), together with the values on the blocks $SM \times N$, $N \times SM$, and then finally resolve the remaining values on the $N \times N$ block. Often, versions of the greedy algorithm are useful for each of these stages of the construction. The precise details are quite technical, particularly for the law $\mathrm{x} \simeq (\mathrm{y} \diamond \mathrm{y}) \diamond ((\mathrm{y} \diamond \mathrm{x}) \diamond \mathrm{y})$ (E1729), which was the last of the equations whose implications were settled by the ETP. We refer the reader to the ETP blueprint for further details.

## 6. Syntactic arguments

Many proofs or refutations of implications (or equivalences) between two equational laws $\mathrm{E}, \mathrm{E}'$ can be obtained from the syntactic form of the equation. We discuss some techniques here that were useful in the ETP.

6.1. **Simple rewrites.** Many equational laws $\mathrm{E}'$ can be formally deduced from a given law E by applying the *Lean* `rw` tactic to rewrite $\mathrm{E}'$ repeatedly by some forward or backward application of E applied to arguments that match some portion of E. For instance, the commutative law E43 clearly implies $\mathrm{x} \diamond (\mathrm{y} \diamond \mathrm{z}) \simeq (\mathrm{y} \diamond \mathrm{z}) \diamond \mathrm{x}$ (E4531) by a single such rewrite. A brute force application of such rewrite methods is already able to directly generate about 15 000 such implications, including many equivalences to the singleton law E2 and the constant law E46. After applying transitive closure, this generates about four million further such implications.

A simple observation that already generates a reasonable number of equivalences is that any equation of the form $\mathrm{x} \simeq f(\mathrm{y}, \mathrm{z}, \dots)$ necessarily is equivalent to the trivial law $\mathrm{x} \simeq \mathrm{y}$, by transitivity; similarly, an equation of the form $f(\mathrm{x}, \mathrm{y}) \simeq g(\mathrm{z}, \mathrm{w}, \dots)$ implies $f(\mathrm{x}, \mathrm{y}) \simeq f(\mathrm{x}', \mathrm{y}')$; and so forth. Equivalences of this form were useful early in the project by cutting down the number of distinct equivalence classes of laws that needed to be studied.

6.2. **Matching invariants.** Fix an alphabet $X$. A *matching invariant* is an assignment $I\colon M_X \to \mathcal{I}$ of an object $I(w) \in \mathcal{I}$ in some space $\mathcal{I}$ to each word $w \in M_X$ with the property that if an equational law $w_1 \simeq w_2$ has matching invariants $I(w_1) = I(w_2)$, then the same